



DUBAI TECHBOOTS LEAGUE SumoX-26

RULEBOOK



CANADIAN
UNIVERSITY DUBAI
Your portal to Canadian education

TABLE OF CONTENT

1. [DEFINITIONS AND TERMINOLOGY](#)
2. [INTRODUCTION TO THE COMPETITION](#)
3. [TEAM ELIGIBILITY AND CONDITIONS](#)
4. [ROBOT SPECIFICATIONS](#)
5. [ARENA SPECIFICATIONS](#)
6. [GENERAL RULES AND REGULATIONS](#)
7. [WARNINGS AND VIOLATIONS](#)
8. [SCORING SYSTEM](#)
9. [ADDITIONAL NOTES](#)



1. DEFINITIONS AND TERMINOLOGY

1.1 ROUND

A Round is a single contest between two teams. It begins with a draw conducted between the official team representatives. The team that loses the draw shall place its robot on the arena first. At the conclusion of the round, **one (1)** Sumo Point shall be awarded to the winning team.

1.2 MATCH

A Match consists of **two (2)** consecutive rounds between the same teams.

If both teams win one round each, a third decisive round shall be conducted to determine the winner of the match.

1.3 VIOLATION

A Violation is considered equivalent to a **red card**.

If a team receives a Violation:

- The team shall be immediately disqualified from the competition.
- All previously earned points shall be forfeited.
- The team shall not be eligible for awards or prizes.

The specific actions that result in a Violation are defined in Section 7 (Warnings and Violations).



1.4 WARNING

A Warning is considered equivalent to a **yellow card**.

It serves as an official notice to the team regarding improper conduct or rule infringement.

If a team accumulates **six (6) warnings**, this shall automatically result in a Violation, leading to disqualification from the competition under the supervision of the organizing committee.

2. INTRODUCTION TO THE COMPETITION

The Sumo Robot Competition is a head-to-head autonomous robotics contest in which two robots, each representing a competing team, compete inside a circular arena with defined specifications (as detailed in later sections of this rulebook). The objective of each robot is to autonomously detect, engage, and push the opponent's robot out of the arena within a designated time limit. All matches are conducted under the supervision of an official judging panel to ensure fairness, safety, and compliance with the competition rules.

Each match between two teams consists of **two (2) rounds**. If both teams win one round each, a third decisive round shall be conducted to determine the match winner.



In exceptional cases, the Chief Judge may authorize an **additional round** in the event of:

- Technical issues affecting the arena or equipment
- Simultaneous exit of both robots resulting in an inconclusive result
- External interference or unforeseen circumstances

Each standard round shall not exceed **three (3) minutes** in duration.

3. TEAM ELIGIBILITY AND CONDITIONS

3.1 ELIGIBILITY

The competition is open to all students currently enrolled in educational institutions within the United Arab Emirates.

3.2 TEAM COMPOSITION

- Each team shall consist of a **minimum of two (2) members** and a **maximum of three (3) members**.
- One member shall be designated as the **Team Leader**, responsible for communicating with the referees and organizing the team during the competition.



3.3 TEAM SUPERVISOR

- Each team may appoint **one supervisor** to oversee the team.
- The designated supervisor cannot be changed during the course of the competition.

3.4 CONDUCT AND SPORTSMANSHIP

- All team members and supervisors are expected to demonstrate the **highest standards of sportsmanship**, integrity, and respect for the rules and regulations of the competition.
 - Any misconduct by a team member or supervisor toward judges, other participants, or staff may result in:
 - A **Warning** for the team, or
 - **Disqualification** from the competition, at the discretion of the organizing committee.
-

4. ROBOT SPECIFICATIONS

4.1 WEIGHT

The total weight of the robot shall **not exceed 3 kilograms (kg)**.



4.2 DIMENSIONS

- The robot's length and width shall **not exceed 20 centimeters (cm)**, including all extensions.
- No specific height limit is imposed.

4.3 SIZE ADJUSTMENT

- The robot is allowed to **change its dimensions after the start of movement**, provided that **all parts remain attached to the central body**.
- Compliance with this rule is determined by the **Match Referee**.

4.4 CONTROL AND AUTONOMY

- **Remote control of the robot is strictly prohibited**; the robot must operate autonomously.
- The primary controller of the robot must be the **new Arduino Uno Q**. The use of any additional microcontrollers or other primary control units is strictly prohibited.
- Teams are free to use any components on their robot – including cameras, motors, sensors, drivers, or other parts—provided that the use of such components does not violate any of the rules or restrictions explicitly stated in this rulebook.



4.5 SENSOR AND INTERFERENCE RULES

- Using devices intended to interfere with the opponent robot's sensors is **prohibited**.
- The front part of the robot, including any blade or diving mechanism, must **not be white or of a color** that could affect sensor readings, as determined by the Match Referee.

4.6 SAFETY AND FAIR PLAY

- The use of any devices or mechanisms that could **cause harm to the arena, the opponent robot, or other participants** is strictly prohibited.
- Storing and deploying **liquids, gases, powders, or projectiles** onto the arena or opponent robot is prohibited.
- The use of **suction devices, vacuum wheels, magnets, or adhesives** to unnaturally fix the robot to the arena surface is prohibited. Compliance is determined by the Match Referee.

4.7 MODIFICATIONS AFTER INSPECTION

- Any modifications or switching of components after official inspection and labeling of the robot may result in disqualification.
- The label with the team name or number must remain attached until the end of the competition.



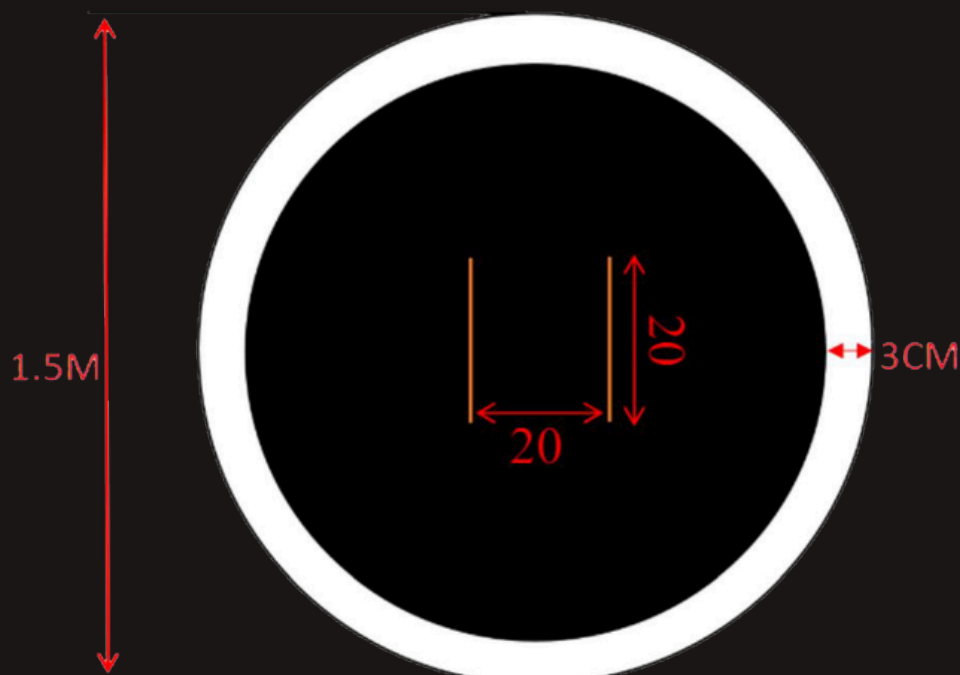
5. ARENA SPECIFICATIONS

5.1 GENERAL STRUCTURE

- The competition arena shall be **circular** and constructed of **wood**.
- The total **diameter of the arena** shall be **1.5 meters (m)**.
- The **arena frame** shall be **white** and have a **thickness of 3 centimeters (cm)**.
- The remaining surface of the arena shall be **black**.

5.2 CENTER MARKINGS

- The arena shall include **two brown lines**, each with a **thickness of 2 cm** and a **length of up to 20 cm**.
- Each line shall be positioned **10 cm from the center** of the arena.



6. GENERAL RULES AND REGULATIONS

6.1 ROUND START

- Each round begins with a **draw between representatives** from the competing teams. The team losing the draw shall place their robot on the arena first.
- Robots may be placed **anywhere along the starting line**, provided they do not touch the line.
- Upon hearing the referee's whistle, the team member shall **activate the robot** and immediately move away to the designated area.

6.2 INITIAL ROBOT MOVEMENT

- After activation, each robot must **remain stationary for at least five (5) seconds** before beginning movement.
- Once both robots start moving and engage to push the opponent, the **referee shall start the official three-minute timer** for the round.

6.3 ROUND AND MATCH SCORING

- A round ends when **one robot successfully pushes the opponent completely out of the arena** within the allotted time



- The winning robot earns **one (1) Sumo Point** for its team.
- A match consists of **two primary rounds**. The team that scores **two points** in these rounds wins the match.
- If both teams win one round each, a **third decisive round** is held.

6.4 MATCH CONCLUSION

- At the end of the match, the referee announces the **winning team and the point totals**.
- Teams must then **remove their robots from the arena and return to the work area**.

6.5 MATCH SCENARIOS

- **Scenario 1:** Team A wins two consecutive rounds → Team B scores zero points.
- **Scenario 2:** Team A and Team B each win one round → Third decisive round determines the winner.

6.6 REFEREE AUTHORITY

The referee has the authority to **stop, suspend, or restart a round** under the following conditions:



A. ARENA DAMAGE:

If a robot damages the arena, the team may receive a **warning** and may **forfeit the round**, depending on the extent of the damage as determined by the referee.

B. TEAM INTERFERENCE:

If any team member approaches or touches a robot or the arena during the match, the team may **lose the round**.

C. Rule Violations:

Any violation of the contest rules may result in the referee stopping the round and penalizing the team.

D. ROBOTS STICKING TOGETHER OR INACTIVE:

If robots **remain stuck together and cease to move for more than thirty (30) seconds**, the referee may **stop the round**.

This includes situations such as:

- One robot mounting the opponent in a way that disables the movement of both robots.
- Robots moving randomly without colliding, resulting in obstruction and disruption of match progress.

E. ROBOT MALFUNCTION OR DAMAGE

- A participating team member may request the suspension of the round if their robot is damaged or malfunctions, preventing it from moving.
- The referee will determine whether the issue is valid; this may result in the team losing the round.



F. TIMEOUTS AND STOPPAGES

- The referee may temporarily stop the round to ensure fair play, verify compliance, or resolve disputes.
- Any stoppage time does not count toward the three-minute match timer.

6.7 MATCH TIMING

- Each standard round shall **not exceed three (3) minutes**.
- The timer **starts immediately after the referee blows the whistle** at the beginning of the round.
- The three-minute timer **excludes the initial five (5) second stationary period** and any time taken by the referee to stop or resume the round.

6.8 TEAM TIMEOUTS

- Each team is allowed **one timeout of up to two (2) minutes per match**.
- No additional timeouts are permitted.

6.9 LATE ARRIVAL

- A team arriving **more than one (1) minute late** for a round shall forfeit that round.



- Teams have **five (5) minutes** to confirm whether they withdraw or continue.
- Being late for **two consecutive rounds** may result in forfeiting the entire match, with the opposing team awarded two points, subject to the referee's discretion.

6.10 TEAM PARTICIPATION

- Only **one team member** may be present to place and activate the robot per round.

6.11 TIE-BREAKER RULES

- If a tie occurs in the **third round** and neither robot pushes the other out:
 - a. The team with the **fewest warnings** is awarded one point.
 - b. If warnings are equal, a **new decisive match** may be conducted at the referee's discretion.
- The referee may also order a replay of the third round if modifications to the robot (programming or placement) are made to ensure fair match progress.

6.12 WARNINGS AND VIOLATIONS

- Accumulating **three warnings in a single round** results in automatic loss of that round.
- Any **robot part detaching** during a round incurs **one warning per detached part**, regardless of size, unless it



disrupts the match progress. The round continues if the match is not affected.

7. WARNINGS AND VIOLATIONS

7.1 VIOLATIONS

- A Violation results in the **immediate disqualification** of the team from the competition.
- Teams receiving a Violation **forfeit all points earned** and become **ineligible for prizes or participation certificates**.
- Accumulating **six (6) warnings** automatically results in a Violation, subject to the discretion of the referees' committee.

7.2 TEAM RESPONSIBILITY

- If a team member receives a warning or violation, the **entire team is considered to have received it**, unless the referee decides otherwise.
- The referee may issue a warning or violation to a specific team member and **disqualify them individually**, provided that the team still has a minimum of **two members** to continue participation.

7.3 CAUSES FOR WARNINGS

A warning may be issued under the following circumstances:



1. **Disrupting the contest or round** – Any action that obstructs the normal progress of the match.
 2. **Unauthorized interference** – Approaching the arena beyond permitted limits or throwing any object that hinders robot movement.
 3. **Premature robot movement** – Activating or moving the robot **before the initial five-second stationary period** after the whistle.
 4. **Unsportsmanlike behavior** – Attempting to distract the referee, opponent, or organizers, or failing to follow the rules and regulations.
 5. **Riotous actions** – Engaging in behavior aimed at disrupting the contest or causing harm to participants or equipment.
-

8. SCORING SYSTEM

- A team is awarded **one (1) Sumo Point** in any of the following cases:
 - When a robot **completely pushes the opposing robot out of the arena**, and any part of the losing robot touches the ground outside the arena.
 - When the opposing robot **exits the arena**, and any part touches the ground, even if there was no direct contact between the robots.



- When the opposing team receives **three (3) warnings** in a single round, the round is considered won by the other team.
- If a team arrives **more than one (1) minute late** for a round, they forfeit that round, and the opposing team receives **one point**.
- A team is declared the **winner of a match** by achieving **two sumo points**, in any of the following cases:
 - Winning **two rounds normally** according to the standard scoring rules.
 - The opposing team **withdraws** or is late for **two consecutive rounds**.
 - The referee determines that the **opponent's robot does not meet specifications** or is ineligible to participate.
 - The opposing team receives a **violation**, resulting in **automatic disqualification**, awarding the match as a **two-point victory**.
 - If a team **intentionally damages the opponent's robot**, the team automatically loses, and the opponent is awarded **two points**.



9. ADDITIONAL NOTES

1. TIE-BREAKING CRITERIA:

- In case of a tie, the following factors will be considered in order:
 - **Clean play:** Teams receiving fewer warnings will have an advantage.
 - **Robot strategy:** The approach and tactics used in attacking the opponent.
 - **Robot movement:** Preference for consistent, deliberate, and non-random movements.
 - **Robot design:** Innovation, efficiency, and overall engineering of the robot.
 - **Player behavior:** Conduct and sportsmanship during matches.

If a tie persists after evaluating all these factors, the referee committee may call for a **decisive match** after allowing sufficient time for adjustments. Should the decisive round still result in a tie, the referee has the authority to **draw lots** to determine the winner.

2. JUDGING DISCRETION:

- Any situation not explicitly covered in this booklet will be resolved at the discretion of the **judging panel**.

3. SAFETY MEASURES:

- A **safety zone of approximately 25–30 cm** will be maintained around the arena. No person is allowed within this distance. Teams will be responsible for any object detected by the sensors beyond this range.

